8 Read the passage below and answer the questions that follow.

In recent years, the global shift towards electric vehicles (EVs) has been driven by the need for sustainable transportation. EVs are often considered 'greener' and more efficient than conventional internal combustion engine (ICE) vehicles. In terms of energy consumption, a typical EV consumes approximately 18 kW h per 100 km, while a conventional petrol vehicle with an ICE consumes about 7.0 litres of fuel per 100 km. Petrol used in ICE vehicles has an energy density of approximately 34.2 MJ per litre.

An EV has a battery that supplies direct current, which an inverter converts into alternating current to drive its electric motor. In the motor, coils of wire are arranged around a rotating part called a rotor and placed in a magnetic field. When current flows through these coils, magnetic forces act on opposite sides of the coil in opposite directions, producing a torque to spin the motor. Fig. 8.1 shows one of the coils.

As the motor spins, a back electromotive force (e.m.f.) is induced in the coils which oppose the supply voltage and plays an important role in regulating the current drawn by the motor.

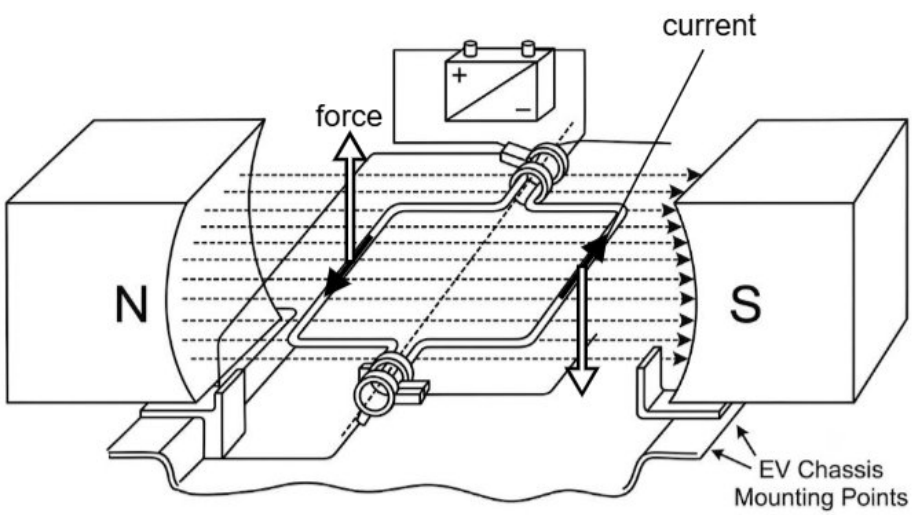


Fig. 8.1

Efficiency in an EV is further enhanced by regenerative braking. When the driver brakes, the motor's role is reversed and it acts as a generator, converting the vehicle's kinetic energy into electrical energy. As chemical batteries cannot absorb these rapid current surges without overheating, ultracapacitors are used as buffers.

Ultracapacitors are energy storage devices with very high capacitance and very low internal resistance. Since an individual ultracapacitor cell has a low voltage rating of 2.8 V, multiple cells are connected in series and parallel to form a high voltage ultracapacitor module needed by an EV.

Engineers tested a prototype ultracapacitor module with internal resistance, by charging it from a completely uncharged state with a constant current of 120 A from a power supply. Fig. 8.2 shows a simple circuit of the connection between the constant current power supply to the prototype ultracapacitor module.

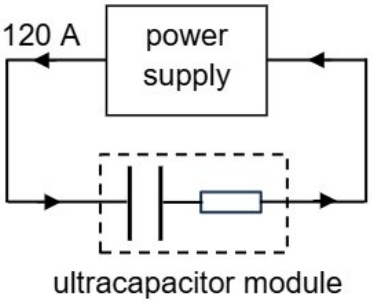


Fig. 8.2

The total voltage V_T across the ultracapacitor module was recorded over time t as shown in Fig. 8.3.

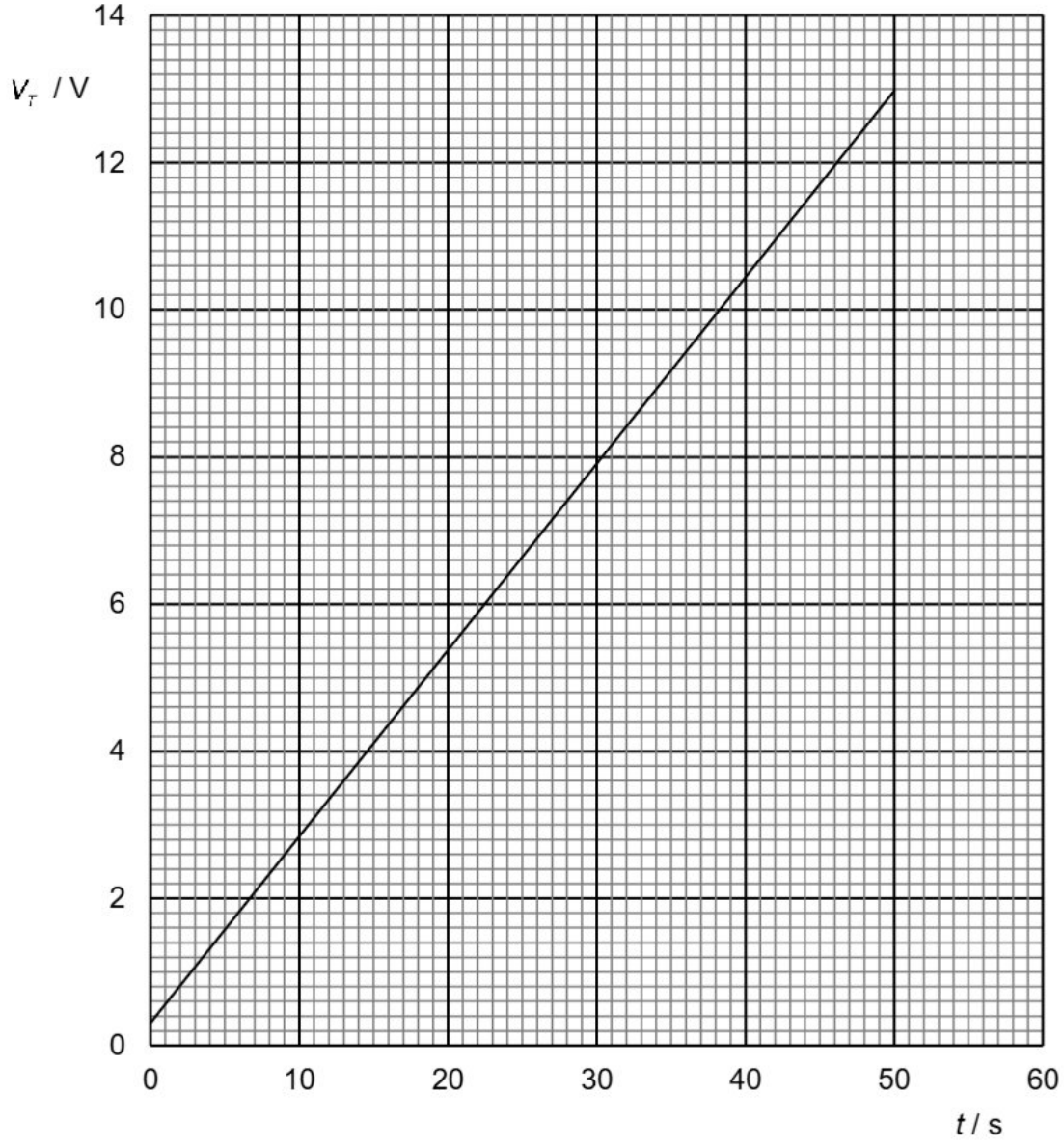


Fig. 8.3

(a) **(i)** For a petrol vehicle, calculate the average power required if the vehicle travels 100 km in one hour at constant speed.

power = ... kW [2]

(ii) The electricity for an EV is generated by a power plant operating at 50 % efficiency.

By comparing the energy per kilometer required by an EV and a petrol vehicle, explain why the EV is still considered 'greener'.

Support your explanation with appropriate calculations.

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..... [2]

(b) As the EV accelerates, the motor spins faster.

(i) Explain why the current in the motor decreases as the motor spins faster.

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..... [2]

(ii) At a certain constant speed, the back e.m.f. is 300 V.

If the supply voltage is 360 V and the total coil resistance is 0.60 Ω , calculate the mechanical power developed by the motor.

power = ... kW [2]

(c) Using Fig. 8.3, determine for the ultracapacitor module,

(i) its internal resistance *r*,

r = ... Ω [1]

(ii) the ideal voltage V_C across it at $t = 40$ s,

V_C = ... V [2]

(iii) its capacitance *C*.

C = ... F [2]

(d) Each individual ultracapacitor cell has a capacitance of 3600 F and a voltage rating of 2.8 V.

(i) Determine the number of cells N that must be connected in series to support a voltage of 42 V across this series string.

N = ... [1]

(ii) Calculate the effective capacitance of N cells in the series string.

capacitance = ... F [1]

(iii) An ultracapacitor module consists of an arrangement of series strings each of N cells.

Compare your answers in **(c)(iii)** and in **(d)(ii)**,

State, with a reason, what can be deduced about the arrangement of the series strings of cells in the module.

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..... [1]

(e) To test an emergency roadside charging system, engineers connected an uncharged ultracapacitor module across a constant 24 V DC power supply in series with the vehicle's climate control system, which acts as a constant resistive load of $0.50\ \Omega$.

Together, the ultracapacitor module and the climate control system form a simple RC charging circuit. The internal resistance of the ultracapacitor module is neglected during this test.

The variation of the voltage V across the ultracapacitor module with time t during this charging process is given by

$$V = V_0\left(1 - e^{-t/\tau}\right)$$

where τ is the time constant of the RC circuit.

Use your value of C from **(c)(iii)** for the ultracapacitor module in the following calculations.

(i) Calculate the value of τ for the emergency charging circuit.

τ = ... s [1]

(ii) On the axes in Fig. 8.4, sketch a graph showing how the voltage V across the ultracapacitor module varies with time t from $t = 0$ to $t = 5\tau$.

Label the numerical value of V at $t = \tau$ and any other key numerical values on the vertical axis.

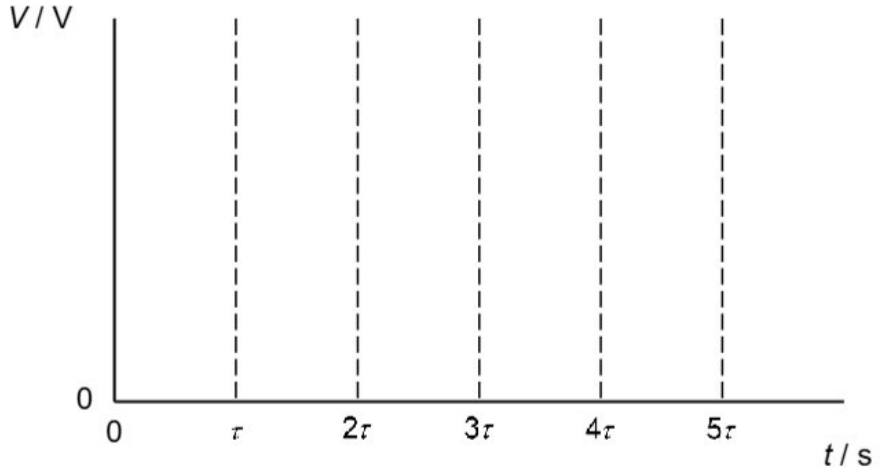


Fig. 8.4

[2]

(iii) The ultracapacitor module is allowed to charge for a long time until it is fully charged to 24 V.

1. Calculate the total electrical energy stored in the fully charged module.

energy = ... J [1]

2. Calculate the total energy delivered by the 24 V DC supply.

energy = ... J [1]

3. State a reason for the difference in the energy values calculated in **(e)(iii)1.** and **(e)(iii)2.**

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..... [1]

[Total: 22]